Lab Values Cheat Sheet

Adult reference intervals, lipid decision limits and drug therapy targets in US conventional and SI units, with conversion factors, traps and nursing cues for NCLEX review

MindShelf

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TRAP Reference intervals vary between laboratories

Interpret a result against the interval printed on **your lab's report**. The values below are one published set, for adults, for study.

Adult lab values

Test	US conventional	Factor	SI
Electrolytes			
Sodium (Na^+)	136–145 mEq/L	× 1	136–145 mmol/L
Potassium (K^+)	3.5–5.0 mEq/L	× 1	3.5–5.0 mmol/L
Chloride (Cl^-)	98–106 mEq/L	× 1	98–106 mmol/L
Bicarbonate (HCO_3^-), serum	23–28 mEq/L	× 1	23–28 mmol/L
Total CO_2, serum	23–30 mEq/L	× 1	23–30 mmol/L
Calcium, total	8.6–10.2 mg/dL	× 0.24951	2.15–2.55 mmol/L
Calcium, ionized	4.65–5.25 mg/dL	× 0.24951	1.16–1.31 mmol/L
Magnesium	1.6–2.6 mg/dL	× 0.4114	0.66–1.07 mmol/L
Phosphorus (as P)	3.0–4.5 mg/dL	× 0.3229	0.97–1.45 mmol/L
Glucose			
Glucose, fasting plasma	70–99 mg/dL	× 0.0555	3.9–5.5 mmol/L
Hemoglobin A1c	4.0–5.6 %	IFCC eq.	20–38 mmol/mol
Kidney			
BUN (urea nitrogen) → urea	8–20 mg/dL	× 0.357	2.9–7.1 mmol/L
Creatinine, female	0.50–1.10 mg/dL	× 88.4	44–97 µmol/L
Creatinine, male	0.70–1.30 mg/dL	× 88.4	62–115 µmol/L
Blood count			
Hemoglobin, female	12–16 g/dL	× 10	120–160 g/L
Blood count			
Hemoglobin, male	14–18 g/dL	× 10	140–180 g/L
Hematocrit, female	37–47 %	× 0.01	0.37–0.47 L/L
Hematocrit, male	42–50 %	× 0.01	0.42–0.50 L/L
WBC (leukocytes)	4,000–11,000 /µL	× 0.001	4.0–11.0 ×10⁹/L
Platelets	150,000–450,000 /µL	× 0.001	150–450 ×10⁹/L
Coagulation			
PT (prothrombin time)	11–13 s	same	11–13 s
INR, no anticoagulant	0.8–1.1	same	0.8–1.1
aPTT	25–35 s	same	25–35 s
Arterial blood gas (room air)			
pH	7.38–7.44	same	7.38–7.44
$PaCO_2$	38–42 mm Hg	× 0.1333	5.1–5.6 kPa
PaO_2	75–100 mm Hg	× 0.1333	10.0–13.3 kPa
HCO_3^-	23–26 mEq/L	× 1	23–26 mmol/L
SaO_2	≥ 95 %	× 0.01	≥ 0.95 (fraction)
Liver panel			
Albumin	3.5–5.5 g/dL	× 10	35–55 g/L
Total protein	5.5–9.0 g/dL	× 10	55–90 g/L
Bilirubin, total	0.3–1.0 mg/dL	× 17.1	5–17 µmol/L
ALT and AST (each)	10–40 U/L	× 0.0167	0.17–0.67 µkat/L
Alkaline phosphatase	30–120 U/L	× 0.0167	0.50–2.00 µkat/L

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Test	US conventional	Factor	SI
Lipids (decision limits)			
Total cholesterol, desirable	< 200 mg/dL	× 0.02586	< 5.17 mmol/L
LDL cholesterol, optimal	< 100 mg/dL	× 0.02586	< 2.59 mmol/L
HDL, female: low if	< 50 mg/dL	× 0.02586	< 1.29 mmol/L
HDL, male: low if	< 40 mg/dL	× 0.02586	< 1.03 mmol/L
Triglycerides, normal	< 150 mg/dL	× 0.01129	< 1.69 mmol/L
Drug therapy (target set by the prescriber)			
INR on warfarin, common	2.0–3.0	same	2.0–3.0
Digoxin, general range	0.5–1.9 ng/mL	× 1.28	0.6–2.4 nmol/L

Ranges vary by lab. SI = US × factor, calculated from the US value except ionized calcium (source in mmol/L). Blood gas: room air.

Convert US ↔ SI

SI = US × factor; US = SI ÷ factor. mg/dL → mmol/L: factor = 10 ÷ molar mass (g/mol). mEq/L → mmol/L: ÷ ion charge (Na^+, K^+, Cl^-, HCO_3^-: same number). mm Hg → kPa: × 0.1333 (1 mm Hg ≈ 133.322 Pa). HbA1c: mmol/mol = (% − 2.152) ÷ 0.09148 (NGSP).

EXAMPLE Worked conversions
Glucose 180 mg/dL × 0.0555 = 9.99 ⇒ **10.0 mmol/L**.
Creatinine 1.5 mg/dL × 88.4 = 132.6 ⇒ **133 µmol/L**.
Ionized Ca 1.10 mmol/L ÷ 0.24951 ≈ 4.4086 ⇒ **4.41 mg/dL**.

Traps

TRAP Same test, two numbers
Glucose 99 mg/dL = 5.5 mmol/L. Read the unit before you compare a result with a range.

TRAP mEq is not always mmol
Equal only for single-charge ions. Ca^{2+}: 1 mmol = 2 mEq; Mg^{2+}: 1 mmol = 2 mEq.

TRAP BUN is not urea
US labs report urea **nitrogen** (BUN); SI labs report urea. Urea has 2 N atoms: × 0.357.

TRAP Total vs ionized calcium
About half of blood calcium is bound to proteins, mainly albumin: abnormal albumin changes **total** calcium, and an ionized (unbound) calcium test may be ordered.

TRAP High K^+ and hemolysis
Report a high K^+ per protocol. Hemolysis (red-cell injury in the sample) can release potassium and falsely raise K^+; a redraw may be requested.

TRAP Hematocrit and SaO_2 in SI
Written as fractions: 37% = 0.37 L/L; 95% = 0.95.

Nursing cues

CLINICAL NOTE INR and bleeding
On warfarin, the prescriber sets the INR target, which depends on the indication; 2.0–3.0 is common. Above 3.0: higher bleeding risk; below 2.0: clot risk. Report per protocol.

CLINICAL NOTE Potassium and the heart
Low K^+ can cause an irregular heartbeat; high K^+ decreases heart muscle activity. Either can be life-threatening: report promptly per protocol.

CLINICAL NOTE Digoxin level
A general range: the right level may vary with the situation. Follow the prescriber's target.

Sources: ABIM Laboratory Test Reference Ranges (Jan. 2026); MedlinePlus; NGSP (HbA1c). Other factors calculated: IUPAC atomic weights, unit definitions, triglycerides as triolein.

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